Axel Gomez

axel.gomez.dev@gmail.com
Axel Gomez AxelGomez16 Web profile
Queretaro, Mexico · Open to Relocation

Professional Summary

Algorithms Engineer with **4+ years in automotive R&D**, building **Python/C++** tools to evaluate perception algorithms, integrate components, and inform release decisions. Experience spans LiDAR ground-truth metrics, parallel algorithm orchestration, and radar-model validation. Pursuing an AI-focused M.S. with research in **hybrid metaheuristic optimization**.

Professional Experience

Aumovio (formerly Continental R&D) June 2022 - Present
Algorithms Engineer *Queretaro, Mexico*

- **Radar Model Validation:** Proposed a **Python** pipeline, then refactored and migrated validation code with **GitHub Actions**, reducing a multi-engineer workflow from approximately **five working days to 50 minutes** for General Motors.
- **Algorithm Orchestration:** Migrated legacy **C++** scheduling for parallel camera/radar algorithms and per-frame failure handling.
- **LiDAR Perception & Evaluation:** Built ground-truth metrics and **SQLite/Dash/Plotly** dashboards for static-object detection and bounding-box query, giving teams shared performance data and reports for Jaguar Land Rover.
- **Perception Integration:** Established two-week checks of combined algorithm performance to inform release decisions; automated regression checks, developed **C++** visualization, and prepared demos including CES.
- **Team Enablement & Code Quality:** Set up Linux environments, trained teammates on internal tools, and helped resolve QAC findings across algorithm components under AUTOSAR/MISRA.

Postealo - Independent Software Product December 2024 - Present
Developer & Maintainer (alongside full-time employment) *Queretaro, Mexico*

- Built and maintain a **Python/LLM** publishing workflow from bot intake to scheduled five-channel delivery; a paying news organization uses it under a **monthly subscription** with ongoing support and feature development.

Continental Talent Entry Program (CTEP) | Systems Engineer Trainee March 2022 - June 2022

- Completed requirements, development, and testing for a door-lock and window-control vehicle prototype before transitioning to algorithms.

Optimization Research (In Progress)

- **Master's thesis - IoT cryptography:** Researching hybrid genetic algorithms and simulated annealing for resource-aware selection of cryptographic algorithms and key lengths; Raspberry Pi evaluation is planned.
- **Bio-inspired optimization:** Collaborating on objective-function optimization for sheep-weight estimation using Grey Wolf Optimizer, Flying Squirrel Optimizer, Cuckoo Search, and Bombardier Beetle Optimizer.

Technical Skills

Programming: Python, C++, C, MATLAB
Optimization (Research): Objective-function optimization, genetic algorithms, simulated annealing, hybrid and bio-inspired metaheuristics
Data & Evaluation: NumPy, Pandas, SQLite, Dash/Plotly, OpenCV, ground-truth metrics
Engineering: Git, GitHub Actions, Linux, CMake, eCAL, QAC, MISRA
Spoken Languages: English, Spanish

Education

Master of Science in Computer Science (Artificial Intelligence) Expected July 2027
Autonomous University of Queretaro *Online, 2025 - Present*

Bachelor of Science in Engineering Physics 2017 - 2021
Autonomous University of Queretaro *Queretaro, Mexico*